THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An echo from 170 m takes (v = 340): 1

(a) 1 s **(b)** 2 s **(c)** 0.1 s **(d)** 0.5 s

2. Speed of sound is highest in: 1

(a) vacuum **(b)** gases **(c)** liquids **(d)** solids

3. A tuning fork produces: 1

(a) heat **(b)** light

(c) sound of a single frequency **(d)** many frequencies

4. Distance covered by sound in 2 s (v = 340): 1

(a) 170 m **(b)** 340 m **(c)** 1360 m **(d)** 680 m

5. The unit of wavelength: 1

(a) metre **(b)** hertz **(c)** joule **(d)** second

6. Stethoscope works on: 1

(a) refraction **(b)** diffraction

(c) multiple reflections **(d)** dispersion

7. Sound travelling 1 km takes about (v = 340): 1

(a) 30 s **(b)** 0.3 s **(c)** 300 s **(d)** 3 s

8. The audible range: 1

(a) 20 kHz to 200 kHz **(b)** 0 to 20 Hz

(c) 2 Hz to 20 Hz **(d)** 20 Hz to 20 kHz

9. Reverberation is reduced by: 1

(a) tiles **(b)** curtains **(c)** bare walls **(d)** glass

10. A compression is a region of: 1

(a) high pressure **(b)** zero pressure **(c)** low pressure **(d)** no particles

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why can sound not travel in vacuum? 2

12. What is amplitude? 2

13. What is loudness? 2

14. Write the formula for speed of a wave. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the wavelength: v = 330 m/s, f = 110 Hz. 3

16. Why do we hear thunder after lightning? Give three points. 3

17. Calculate the distance covered by sound in 3 s at 340 m/s. 3

OR

Find the frequency: v = 340 m/s, lambda = 0.5 m. 3

18. Name the three types of mediums with the speed of sound in each (approx). 3

OR

Why are echoes not heard in a small room? Give three points. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how SONAR works. 5

OR

Compare the speed of sound in different media. 5

20. A man claps and hears an echo after 2 s. Find the distance of the wall (v = 340). 5

OR

Explain noise pollution. 5

END OF QUESTION PAPER Total: 40 Marks

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